AP PRECALCULUS

Unit 1 • Worksheet 1.3B

Worksheet 1.3B • Rates of Change in Linear and Quadratic Functions

Name: _____ Date: _____

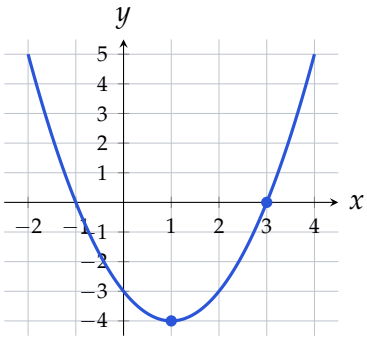
Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

The graph of a quadratic f is shown. What is the average rate of change of f on $[1, 3]$?



(A) -2

(B) 4

(C) 2

(D) 0

Question 2

NO CALCULATOR

For the same function, the average rate of change on $[-1, 1]$ is -2 and on $[1, 3]$ is 2. Which statement explains this?

(A) f is linear.

(B) The rate of change of f is increasing, so the graph is concave up.

(C) The rate of change of f is decreasing.

(D) f has a maximum at $x = 1$.

Question 3

NO CALCULATOR

At what value of x does the rate of change of $f(x) = 3x^2 - 12x + 1$ change from negative to positive?

(A) $x = 2$
(B) $x = -2$
(C) $x = 4$
(D) $x = 12$

Question 4 (Free response)

NO CALCULATOR

The height of a thrown ball is recorded each second: $h(0) = 2$, $h(1) = 17$, $h(2) = 22$, $h(3) = 17$, $h(4) = 2$ (meters).

(a) Find the average rates of change on $[0,1]$, $[1,2]$, $[2,3]$, $[3,4]$. What do you notice?
(b) Write a quadratic model $h(t)$.
(c) Is the graph of h concave up or concave down? Interpret the rate of change in context.